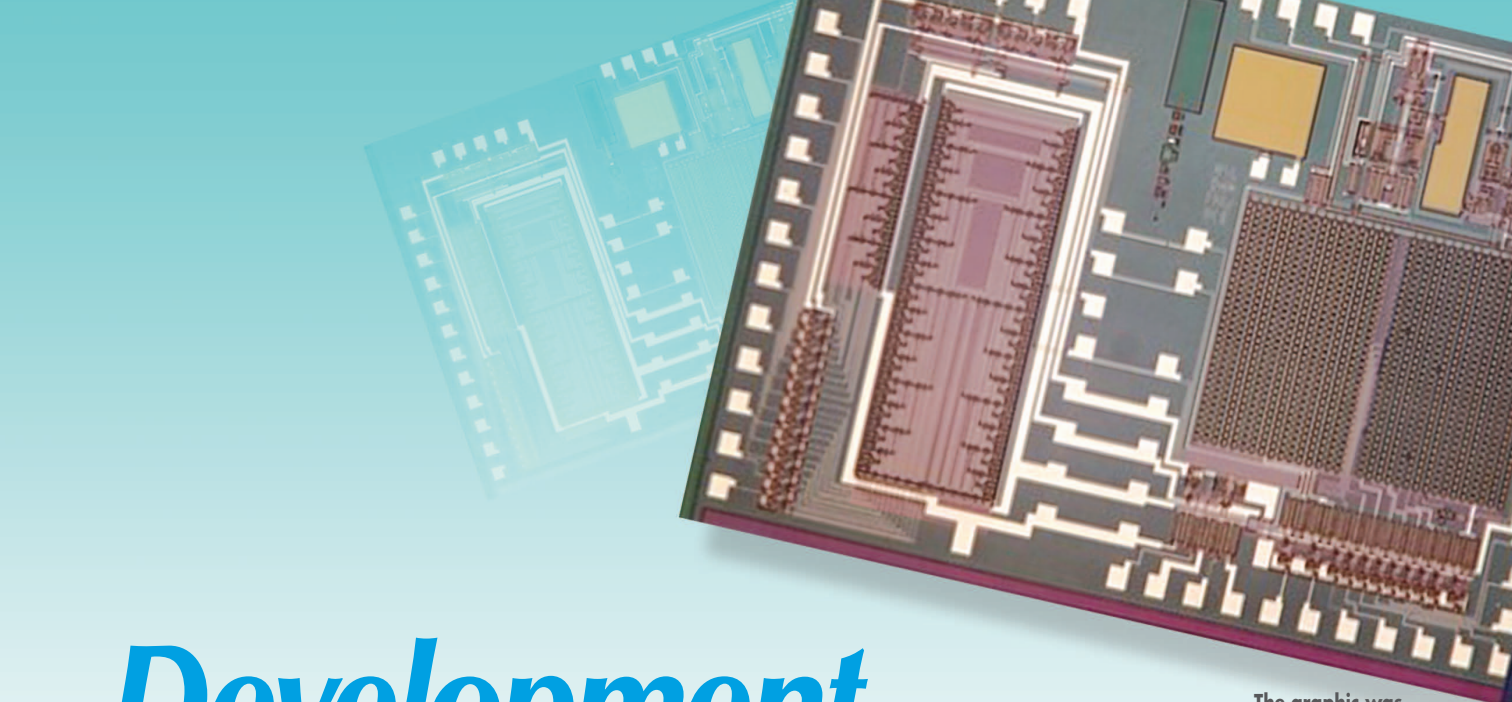




Development of Self-Calibrating A/D Converters

The graphic was generated from an image of an ADC by Lee, Hodges, and Gray (Figure 3).

Introduction

The data converters – analog-to-digital and digital-to-analog converters (ADC's and DAC's respectively) – provide an important interface function between analog and digital circuits in most electronic systems. As MOS integrated circuit technologies began to replace bipolar technologies in the 1960s and 1970s, it became realistic to integrate both complicated digital and high performance analog circuits on a single silicon die. Therefore, there emerged a strong need for MOS data converters. The most popular data converter architecture for bipolar integrated circuits was an R-2R ladder and its variants. However, the lack of precision resistors and low transconductance in MOS transistors made the R-2R ladder difficult to provide comparable accuracy in MOS technologies.

For analog circuit applications, the benefits of MOS technology included the availability of high quality capacitors, excellent switching characteristics, and the zero gate current. The weakness were the aforementioned resistor accuracy,

low transconductance, and lower power supply voltages. Radically different circuit architectures that exploited the benefits while avoiding the drawbacks of MOS technology were needed. The switched-capacitor filters were a great example of such architecture [1]. In the similar line of thoughts, James McCreary along with Profs. Paul Gray and David Hodges, came up with an innovative ADC architecture using switched capacitors [2]. Rather than employing a separate sample-and-hold amplifier (SHA), which was a common practice in successive approximation ADC's, the input was sampled directly on the binary weighted capacitor array. Not only this eliminated the need for a separate SHA with associated power, noise, and area, it also removed the linearity degradation due to the nonlinear parasitic capacitance at the input of the comparator. The result was 10-bit accuracy primarily limited by the capacitor ratio matching. This technique referred to as charge-redistribution ADC would become the most dominant successive approximation ADC architecture for the next 4 decades.

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Self-Calibrating Data Converters

Although the charge-redistribution ADCs revolutionized MOS data converters, the accuracy was limited to 10 bits or lower. Through the 1970s, high-precision (≥ 12 bits), high-speed (≥ 1 kS/s) data converters were predominantly built in ‘hybrid’ technologies in which multiple components are packaged in a relatively large enclosure. The core component that determined the accuracy was an R-2R ladder DAC where the metal thin-film, typically NiCr, resistors were factory laser-trimmed to the desired accuracy against a golden standard –accurately calibrated input voltages. Obviously, such data converters were expensive due to the individual laser trimming, and bulky due to hybrid packaging. As an example, Analog Devices introduced a family of then state-of-the-art DAS11xx series ADCs in 1982. The DAS1140 was a 16b 30 kS/s ADC without a SHA, and the DAS1153 a 15b 30 kS/s ADC with a SHA [3]. The DAS1140 had dimensions of 51 mm x 51 mm x 11.2 mm, and a volume price of \$149, which in today’s dollars is \$360. At the end of 1970s, dynamic component matching technique surfaced, which enabled high accuracy without laser trimming [4]. However, due to a large supply voltage requirement for stacked current dividers, this technique was not generally compatible with MOS technologies.

The challenge that Prof. David Hodges put on my doctoral research in June of 1981 when I joined his group was to come up with a viable MOS alternative to such beasts. This would then enable cohabitation of high performance digital and analog circuits on a single die. The initial goal was 16-bit accuracy, which was a very tall order at the time. Although the ingenious capacitor array charge-redistribution ADC had opened an era of MOS data converters [2], the accuracy, mainly determined by capacitor matching, had been limited to 10 bits. Bahram Fotouhi and Prof. Hodges had developed a MOS 12-bit ADC by a clever combination of resistor-ladder main-DAC and a charge-redistribution

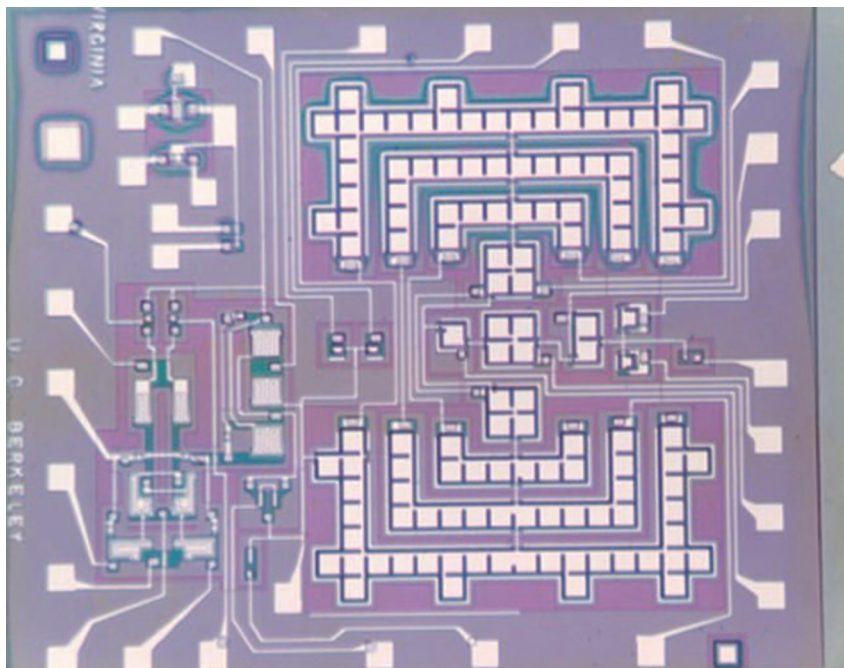


FIGURE 1: The MOS charge-redistribution ADC [2]. It samples the input directly on the weighted capacitor array eliminating a SHA and nonlinearity due to parasitic capacitance. 10 bit accuracy was achieved.

sub-DAC [5]. While 12-bit monotonicity was achieved, the accuracy (integral linearity or INL) was actually limited to 6 bits due to the poor matching characteristic of the diffused resistors in the main DAC. Therefore, the difficulty was how to achieve high accuracy given 6 bits

(resistors) and 10 bits (capacitors) of matching accuracy. Laser trimming was a non-starter because the primary driver for MOS technologies is the high level of integration and low cost. Furthermore, standard MOS technologies were incompatible with laser trimming, anyway.



FIGURE 2: The 15 and 16 bit hybrid ADC’s from Analog Devices, Inc. introduced in 1982. They were 102mm x 102mm x 11.2mm and 51mm x 51mm x 11.2mm in size [3].

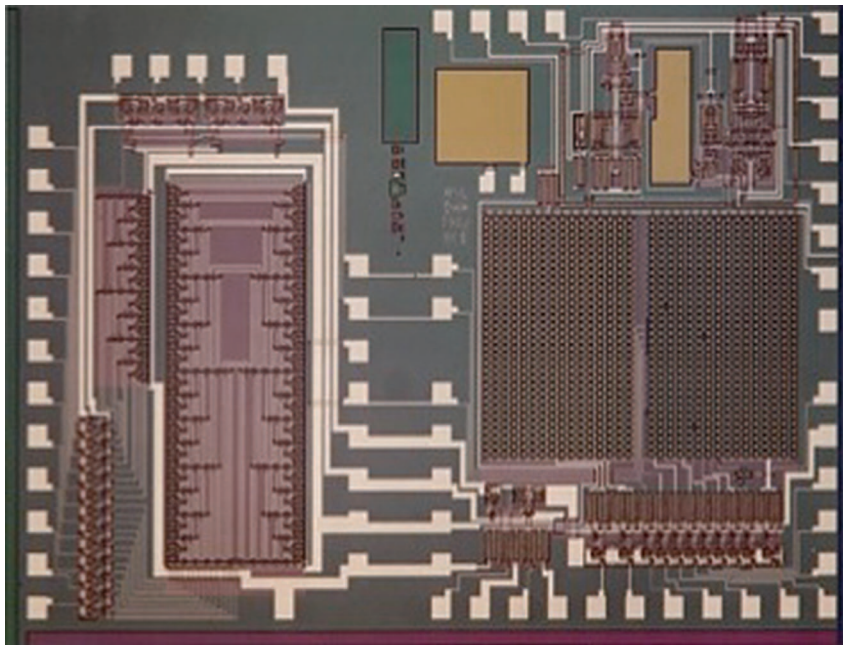


FIGURE 3: Self-calibrating ADC, consisting of a 10 bit capacitor array and 2 resistor ladder sub-DAC and calibration-DAC's [7]. Due to the possible yield concerns, the capacitor array and the resistor ladders were laid out so that they could be separated and stitch-bonded if necessary.

An interesting 14-bit MOS ADC was presented at the 1981 ISSCC by Ziya Boyacigiller, et al. from Intersil, Inc. [6]. It was an R-2R ladder-based ADC where the effective bit weights were deliberately made less than 2, nominally 1.85, by exercising the 4th bit down along with the present bit during the successive approximation. The resulting sub-radix-2 algorithm provided enough redundancy to ensure the matching errors could be compensated in the digital domain. The actual bit weights were factory-measured using accurately calibrated input voltages (golden standard) and stored in an EPROM. The stored true bit values were accessed during the normal conversion for an accurate result. Although redundancy and digital correction is common nowadays, this is the earliest such example that I could identify. It was a pioneering technique that eliminated the need for laser trimming, but many of the key issues remain unsolved – the need for individual factory calibration with a golden standard and the need for non-volatile memory, which was incompatible with standard MOS technologies.

Thus started my quest for a method to compensate for component matching errors without individual factory

calibration. While walking on Berkeley campus about 3 weeks after the 16-bit challenge was given, an idea hit me that set a chill along my spine. I realized that the binary weighted capacitor array had a very interesting property – the nominal value of any bit capacitor was equal to the sum of all lower bit capacitors. Thus, by comparing a capacitor with the sum of lower capacitors, the matching error could be measured without a golden standard. The 16-bit ADC structure would employ a 10-bit capacitor array main-DAC and 6-bit resistor ladder sub-DAC, each of which was required to be accurate to its own resolution only. This was consistent with the matching characteristics of capacitors and resistors at the time. Note the main-DAC and sub-DAC choices were the opposite of the Fotouhi design. In the new architecture, it was the capacitor array main-DAC that determined the overall accuracy, thus the uncalibrated accuracy would have been limited to 10 bits. The converter would have a separate ‘calibration procedure’ which would be initiated at power-up. Starting with the MSB capacitor, the reference voltage was sampled on it and then the charge was sloshed to the lower capacitors. The resulting

small change of voltage on the capacitor array was a direct measure of the MSB capacitor mismatch error, which could be accurately measured by a normal successive approximation of the sub-DAC (in the final implementation, another 7-bit ‘calibration’ DAC was included for better results). After the errors were measured and stored in a RAM, the converter was ready for a normal conversion. When each bit was exercised in successive approximation, the digitally stored error was accessed, put through the calibration DAC to precisely cancel the capacitor mismatch error.

I ran behavioral simulations on Fortran, and convinced myself this was the breakthrough that I was looking for. Excited, I made a presentation to Prof. Hodges, which was met with the initial skepticism, “How is it fundamentally possible to calibrate itself with out a golden-standard, there can’t be enough number of equations for the unknowns?” That question required a quite a bit of thought to answer, but the fact was that the capacitor ratios were all that mattered, not the absolute values, and the proposed technique measured ratio errors. Once that doubt was erased, Prof. Hodges was enthused, and we made a presentation to Prof. Paul Gray to make sure that we were not missing anything, and he also agreed that it was a great idea. It was then he coined the term ‘self-calibration’. The ensuing transistor-level design went smoothly albeit with some challenges due to the high level of the targeted accuracy.

One major issue was access to fabrication. At that time, there were no foundry services available to submit the design for fabrication. The graduate students had to fabricate their own chips in the laboratory from scratch all the way from mask generation, lithography, etching, diffusion, to device characterization. The standard Berkeley CMOS technology was a 10 V, 1-metal layer, 5 μ process. As usual for a student without prior fabrication experience, mistakes were made, and several runs had to be scrapped before the first functional batch was

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produced after about 6 months of hard labor in the laboratory. Although fully functional, the accuracy was disappointing 12-bits. Hundreds of different tests were performed in an attempt to improve the accuracy with no avail. Some sort of a hysteresis-like behavior was observed in the comparator, which I attributed to traps due to contamination at silicon-silicon dioxide interface. Another fabrication run with a special care in clean gate oxide growth resulted in no improvement. Yet another fabrication run, this time in the brand new lab, new equipment, and new process flows to improve the interface quality did nothing to improve the accuracy at initial test. However, there was one important distinction in this batch – slowing the clock frequency significantly improved the accuracy even though the previous batches showed no improvement. Although designed for 50 kS/s rate, when run at 12 kS/s, it achieved 15-bit accuracy, which was deemed the best achievable given that the capacitor array matching was 1 bit short of expectation [7]. The hysteresis-like behavior in the comparator led to later research at MIT on oxide traps and electron and hole tunneling that theoretically and experimentally explained the observed behavior [8]. The reason why the accuracy of the last batch improved at lower speed, I believe, was because it had different trap density versus depth profile. Apparently, the lower density of deeper traps in the new batch allowed relaxation of hysteresis at slower clock frequencies.

Epilog and Conclusion

Until well after I thought of the self-calibration idea, I believed I was the first one to have recognized the property of the binary weighted capacitor array and exploited it to measure capacitor mismatch – nothing remotely similar

had turned up in any of the circuit-type journal or conference publications. I discovered later, to my dismay, McCreary had thought of the same principle and had applied it to characterize capacitor mismatch characteristics [9]. This had been published in *IEEE Transactions on Instrumentation and Measurements* in 1979 – a journal I had not thought was relevant to data converters. Of course, McCreary used the principle for capacitor matching characterization, not for calibrating ADC's, but he recognized the important symmetry in a binary weighted capacitor array and used it to measure capacitor mismatch. Along with his pioneering work in charge-redistribution ADC's, his work laid the foundation for the self-calibrating ADC work presented here.

The self-calibrating ADC improved its own accuracy without a factory calibration or a golden standard for the first time. Many other calibration methods followed for various different types of ADC's including successive-approximation, pipeline, and delta-sigma ADC's, ultimately finding applications in other analog circuits in the form of now popular digitally-assisted analog circuits.

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